

18.600: Lecture 39

Review: practice problems

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Markov chains

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- ▶ When Alice showers, she first checks to see if at least one towel is present. If a towel is present, she dries off with that towel and returns it to the bathroom towel rack. Otherwise, she cheerfully retrieves both towels from the walk-in closet, then showers, dries off and leaves both towels on the rack.

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- ▶ When Alice showers, she first checks to see if at least one towel is present. If a towel is present, she dries off with that towel and returns it to the bathroom towel rack. Otherwise, she cheerfully retrieves both towels from the walk-in closet, then showers, dries off and leaves both towels on the rack.
- ▶ **Problem:** describe towel-distribution evolution as a Markov chain and determine (over the long term) on what fraction of days Bob emerges from the shower to find no towel.

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$$M = \begin{pmatrix} 0 & .5 & .5 \\ .5 & 0 & .5 \\ 0 & 1 & 0 \end{pmatrix}$$

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- ▶ Row vector π such that $\pi M = \pi$ (with components of π summing to one) is $(\frac{2}{9} \quad \frac{4}{9} \quad \frac{1}{9})$.
- ▶ Bob finds no towel only if morning starts in state zero and Bob goes first. Over long term Bob finds no towel $\frac{2}{9} \times \frac{1}{2} = \frac{1}{9}$ fraction of the time.

Optional stopping, martingales, central limit theorem

Suppose that $X_1, X_2, X_3, \dots$ is an infinite sequence of independent random variables which are each equal to 1 with probability $1/2$ and -1 with probability $1/2$. Let $Y_n = \sum_{i=1}^n X_i$. Answer the following:

- ▶ What is the probability that Y_n reaches -25 before the first time that it reaches 5 ?

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- ▶ What is the the probability that Y_n reaches -25 before the first time that it reaches 5 ?
- ▶ Use the central limit theorem to approximate the probability that $Y_{9000000}$ is greater than 6000 .

Optional stopping, martingales, central limit theorem — answers

- ▶ $p_{-25}25 + p_55 = 0$ and $p_{-25} + p_5 = 1$. Solving, we obtain $p_{-25} = 1/6$ and $p_5 = 5/6$.

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- ▶ $p_{-25}25 + p_55 = 0$ and $p_{-25} + p_5 = 1$. Solving, we obtain $p_{-25} = 1/6$ and $p_5 = 5/6$.
- ▶ One standard deviation is $\sqrt{9000000} = 3000$. We want probability to be 2 standard deviations above mean. Should be about $\int_2^\infty \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$.

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- ▶ Yes, no, yes, no.

Calculations like those needed for Black-Scholes derivation

- ▶ Let X be a normal random variable with mean 0 and variance 1. Compute the following (you may use the function $\Phi(a) := \int_{-\infty}^a \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$ in your answers):

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 - ▶ $E[e^{3X-3}]$.
 - ▶ $E[e^X 1_{X \in (a,b)}]$ for fixed constants $a < b$.

Calculations like those needed for Black-Scholes derivation

– answers

$$\begin{aligned} E[e^{3X-3}] &= \int_{-\infty}^{\infty} e^{3x-3} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2-6x+6}{2}} dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2-6x+9}{2}} e^{3/2} dx \\ &= e^{3/2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-3)^2}{2}} dx \\ &= e^{3/2} \end{aligned}$$

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$$\begin{aligned} E[e^X 1_{X \in (a,b)}] &= \int_a^b e^x \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &= \int_a^b e^x \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &= \int_a^b \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2 - 2x + 1 - 1}{2}} dx \\ &= e^{1/2} \int_a^b \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-1)^2}{2}} dx \\ &= e^{1/2} \int_{a-1}^{b-1} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &= e^{1/2} (\Phi(b-1) - \Phi(a-1)) \end{aligned}$$

If you want *more* probability and statistics...

► **UNDERGRADUATE:**

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► OUTSIDE OF MATH DEPARTMENT

- (a) Look up new MIT minor in statistics and data sciences.
- (b) Look up *long* list of probability/statistics courses (about 78 total) at <https://stat.mit.edu/academics/subjects/>
- (b) Ask other MIT faculty how they use probability and statistics in their research.

Thanks for taking the course!

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- ▶ Happy exam day, May 17!
- ▶ And may the odds be ever in your favor.